

CareConnect: Assistance & Social Connection Platform



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Abstract

CareConnect is a web-based platform that connects individuals needing daily assistance with trusted service providers called CareMates. Users can request help, set reminders, update mood, and engage socially. CareMates accept tasks based on urgency and availability. The system ensures accessibility, reliability, and community support for all age groups.

Problem Statement

- Many individuals, especially the elderly, lack reliable access to daily assistance for tasks like transportation, shopping, or medical needs.
- Existing service platforms often lack urgency-based task assignment, mood tracking, and reminders.
- There is limited digital infrastructure to connect verified helpers with users in a safe, location-aware, and socially engaging environment.

Objectives

- To address urgent real-life help needs without a structured system.
- To develop a single platform for tasks, social interaction, and health tracking.
- To promote a community-driven support model with paid professionals (CareMates).

Literature Review

Software	Features	Limitations
Care.com	Hire caregivers for various tasks.	No real-time task urgency system.
AgingCare	Provides resources & articles for elderly care.	Lacks interactive features (mood tracking, social board).
Papa	Offers elderly companionship & assistance.	Limited task categories (no medication reminders, social board).
CareConnect (Ours)	Task requests, mood tracking, CareMate matching, community board, health reminders.	New project- needs adoption & scaling.

Fig-1: Literature Review.

Software Requirements

Functional Requirements:

- Help Request
- Set Reminders
- Social Posts
- Mood Update
- Payment Integration

Non-Functional Requirements:

- Usability
- Performance
- Reliability
- Scalability
- Security

High-Level Architecture

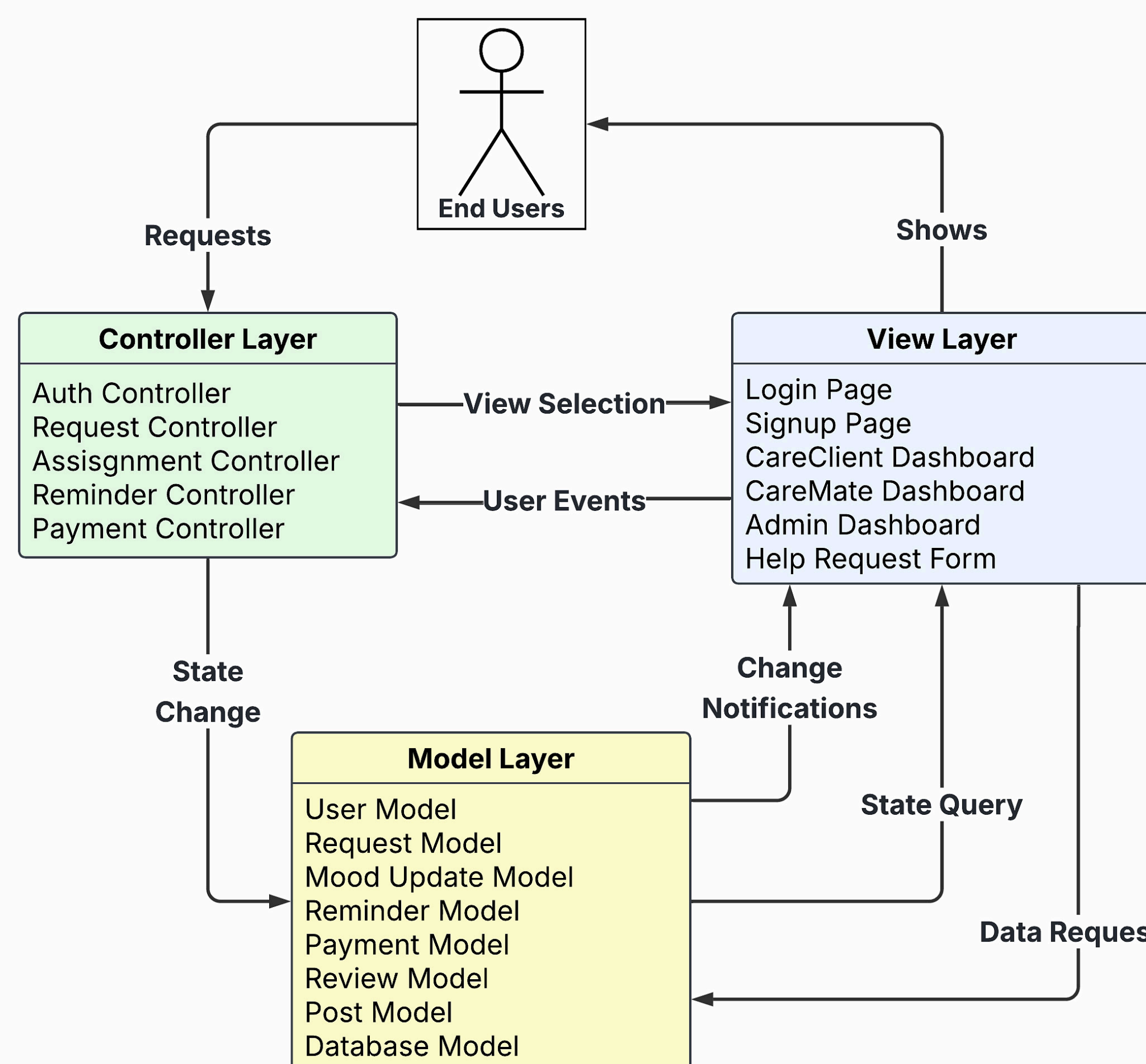


Fig-2: High-Level Architecture of CareConnect.

Low-Level Architecture

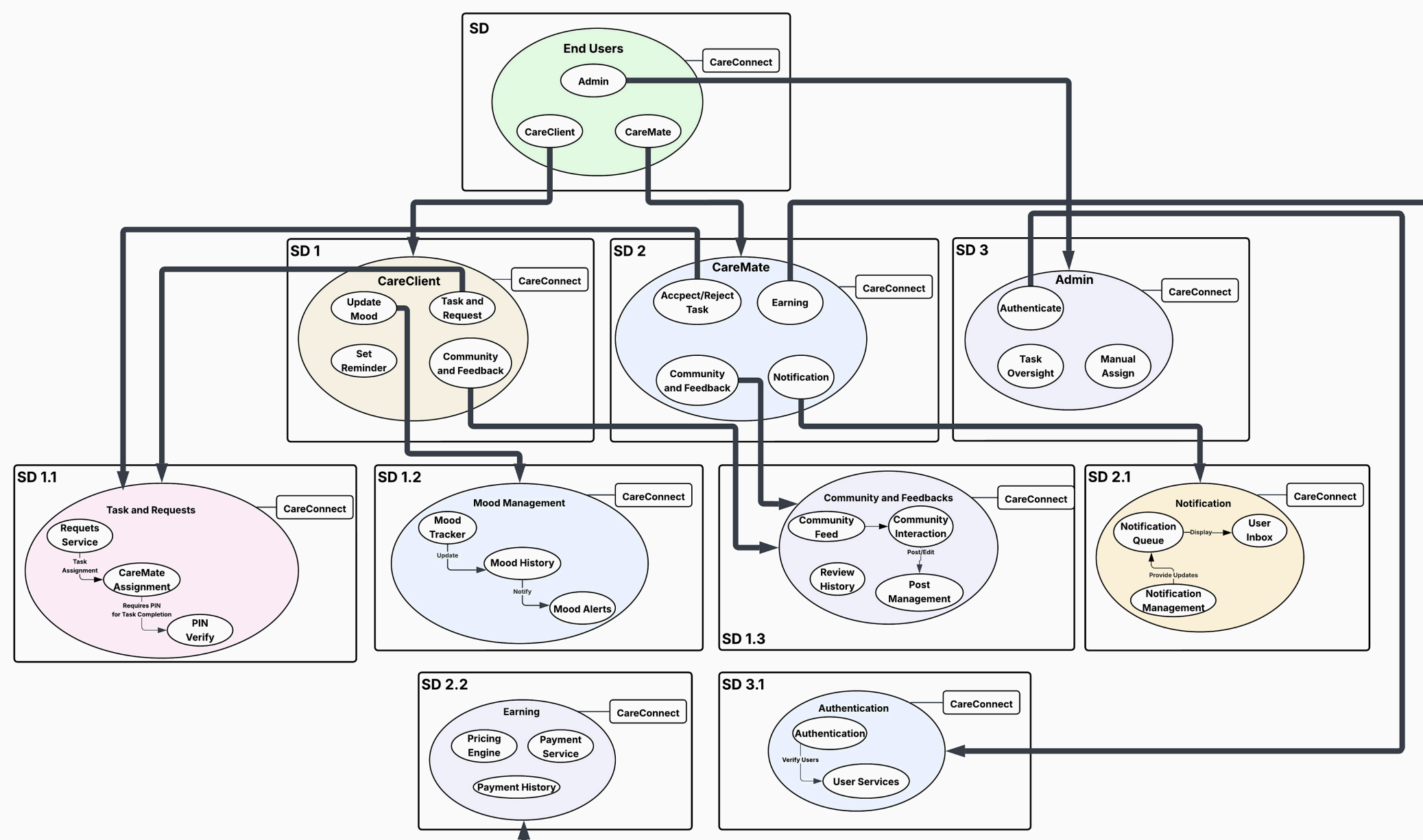


Fig-3: Low-Level Architecture of CareConnect.

ER Diagram

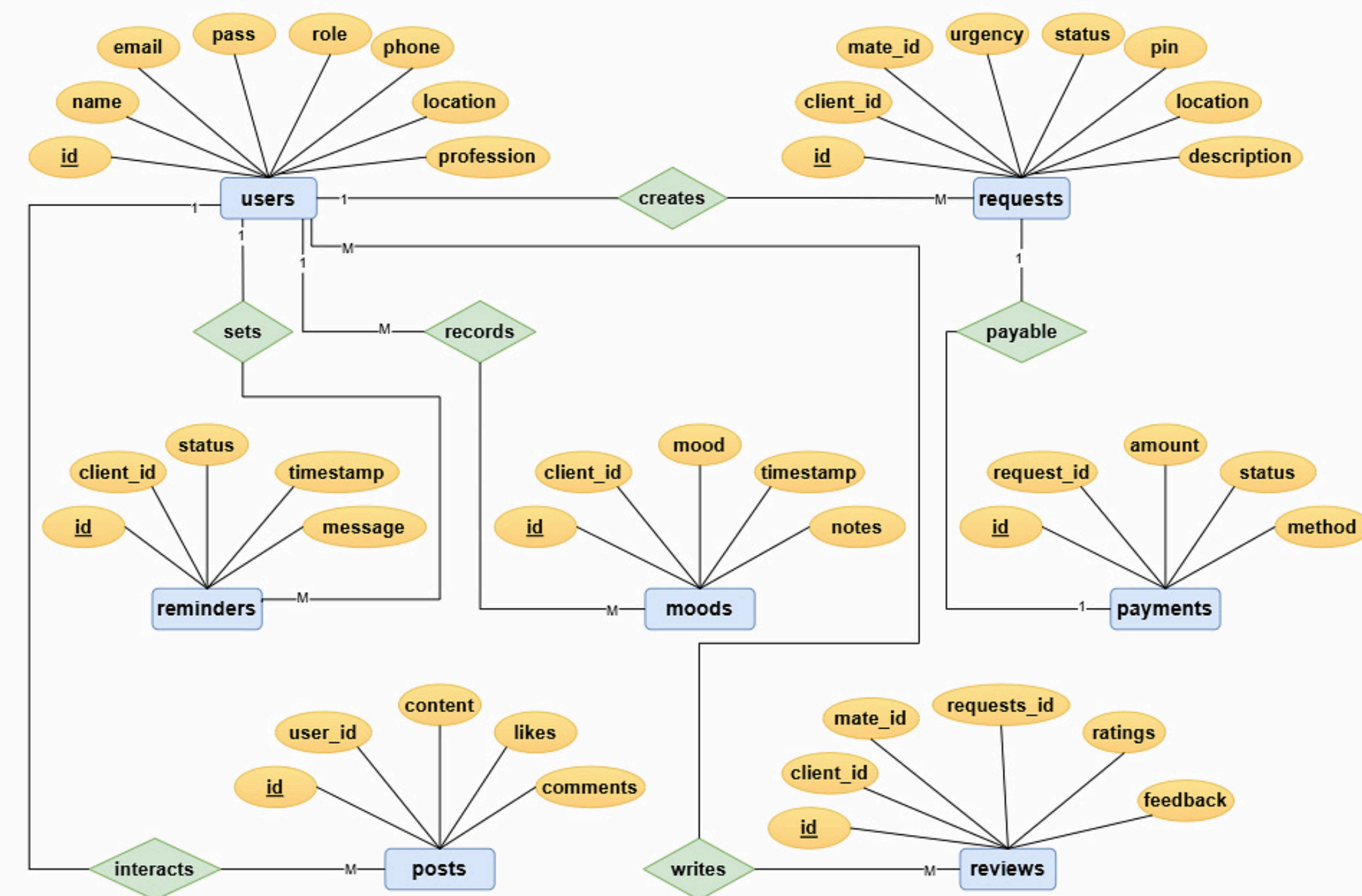


Fig-4: ER Diagram of CareConnect.

Test Cases

Authentication Module						
TID	Test Scenario	Pre-Condition	Input/Test Case	Expected Output	Actual Output	Test Result
AUTH-01	User login in successfully	User already registered	Valid mail + password	Redirect to dashboard	As expected	Pass
AUTH-02	Invalid login attempt	User registered	Wrong password	Error: "Incorrect password or email"	As expected	Pass
AUTH-03	Login with empty fields	None	Blank email/password	Error: "Fields cannot be empty"	As expected	Pass
AUTH-04	Successful Signup	User not registered	Name, email, password, role	Account Created	As expected	Pass
AUTH-05	Duplicate Signup	Email already exists	Using existing email	Error: "Email already registered"	As expected	Pass
AUTH-06	Role-based login	Role selected (Client/Caremate/Admin)	Valid credentials	Redirect to respective dashboard	As expected	Pass
Help Request Module						
TID	Test Scenario	Pre-Condition	Input/Test Case	Expected Output	Actual Output	Test Result
REQ-01	Create a help request	CareClient logged in	Request details + urgency	Request saved + PIN generated	As expected	Pass
REQ-02	Missing Fields	CareClient logged in	Empty description/location	Validation error	As expected	Pass
REQ-03	Create recurring request	CareClient logged in	Request + recurring type	Recurring request	As expected	Pass
REQ-04	View created requests	CareClient logged in	Click "My Request"	List of requests displayed	As expected	Pass
REQ-05	View assigned CareMate	CareClient logged in	Open request details	Open request details	As expected	Pass
CareMate Task Management Module						
TID	Test Scenario	Pre-Condition	Input/Test Case	Expected Output	Actual Output	Test Result
CM-01	View available tasks	CareMate logged in	Open task list	Pending tasks displayed	As expected	Pass
CM-02	Accept a task	Task available	Accept task	Task assigned to CareMate	As expected	Pass
CM-03	Reject a task	Task available	Reject task	Task returns to pool	As expected	Pass
CM-04	Complete task successfully	Assigned task + correct PIN	Enter valid PIN	Task Completed + payment processed	As expected	Pass
CM-05	Wrong PIN during completion	Assigned task	Enter wrong PIN	"Invalid PIN" error	As expected	Pass
CM-06	View assigned tasks	CareMate logged in	Click "My Tasks"	Assigned task list displayed	As expected	Pass
Reminder Module						
TID	Test Scenario	Pre-Condition	Input/Test Case	Expected Output	Actual Output	Test Result
REM-01	Set a reminder	CareClient logged in	Reminder details	Reminder saved	As expected	Pass
REM-02	Update reminder	Reminder exists	Edit reminder	Reminder updated	As expected	Pass
REM-03	Reminder triggers notification	Reminder time reached	Wait for time	Notification received	As expected	Pass
REM-04	Delete reminder	Reminder exists	Delete reminder	Reminder removed	As expected	Pass
Mood Update Module						
TID	Test Scenario	Pre-Condition	Input/Test Case	Expected Output	Actual Output	Test Result
MOOD-01	Update mood	CareClient logged in	Select daily mood	Mood saved	As expected	Pass
MOOD-02	View mood history	Previous moods exist	Open history	Mood timeline displayed	As expected	Pass
MOOD-03	Trigger 24-hour alert	No mood update in 24h	Wait 24h	CareMate received notification	As expected	Pass
MOOD-04	Check mood alert delivery	Mood not updated	Trigger alert manually	Notification delivered	As expected	Pass
Social Community Module						
TID	Test Scenario	Pre-Condition	Input/Test Case	Expected Output	Actual Output	Test Result
SOC-01	Create a post	CareClient logged in	Type text post	Post displayed	As expected	Pass
SOC-02	Empty post	CareClient logged in	Submit empty	Error shown	As expected	Pass
SOC-03	View all posts	Posts exist	Open board	All posts displayed	As expected	Pass
SOC-04	Like/Comment on posts	Posts exist	Like/Comment	Interaction saved	As expected	Pass
Admin Module						
TID	Test Scenario	Pre-Condition	Input/Test Case	Expected Output	Actual Output	Test Result
ADM-01	View all Users	Admin logged in	Open user list	User list displayed	As expected	Pass
ADM-02	Manually assign CareMate	Task exist	Select CareMate	Assignment Complete	As expected	Pass
ADM-03	View Logs	Log exist	Open Logs page	User logs displayed	As expected	Pass
ADM-04	Remove Users	User exists	Delete User	User removed	As expected	Pass

Fig-5: Tested Cases of CareConnect.

Tools and Technologies

- Frontend: HTML, Tailwind CSS, JavaScript (ES6)
- Backend: PHP, AJAX
- Database: MySQL
- Development Tools: VS Code, Cursor, GitHub
- Testing & Debugging: Browser DevTools, PHP Error Reporting
- Deployment & Environment: Apache Server, Localhost Configuration (XAMPP)

System Overview

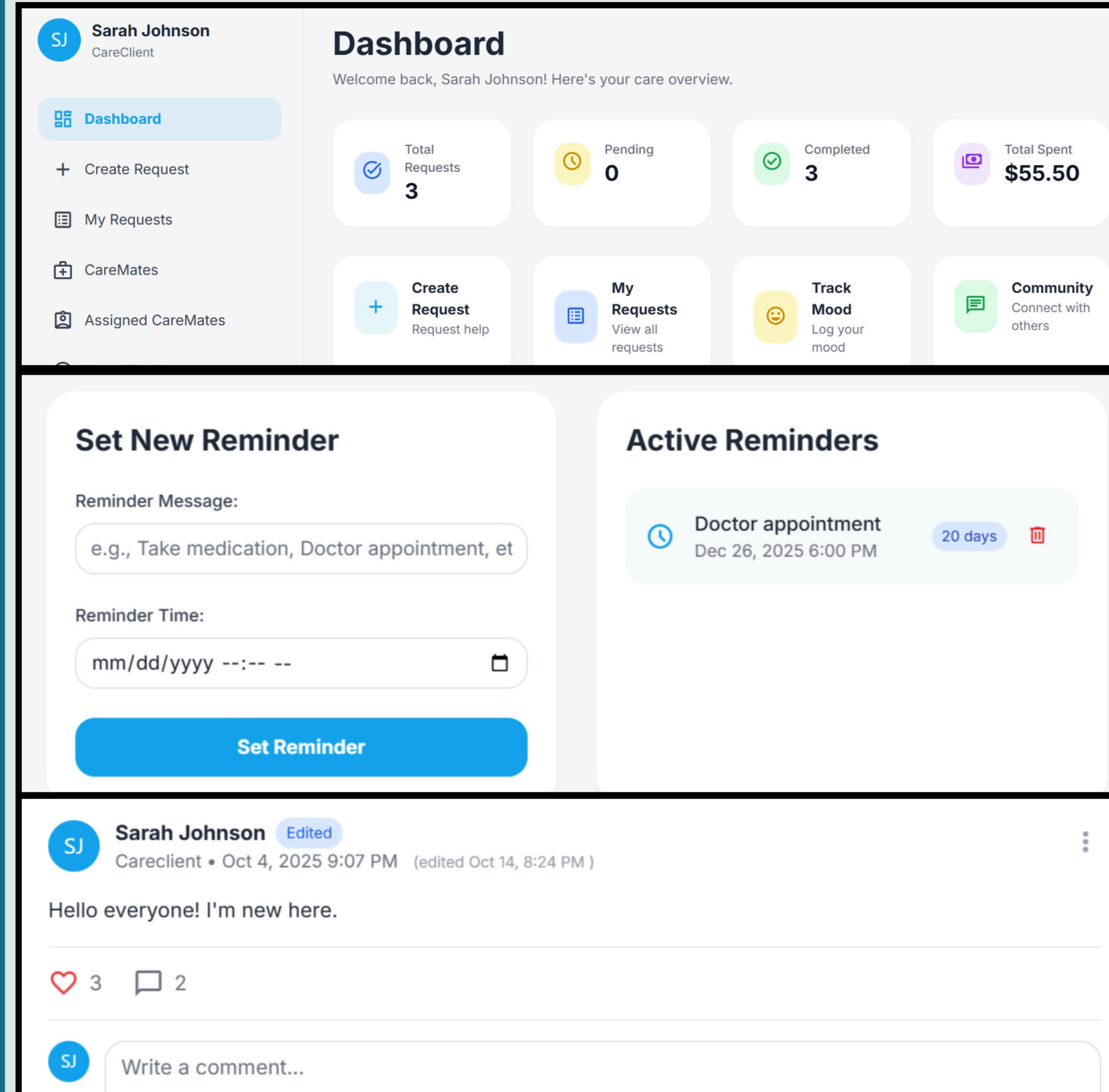


Fig-6: Working Overview of CareConnect.

Result and Validation

All Functional Requirements Met

- Help Request, Reminders, Social Posts, Mood Updates, and Payment flow are fully operational.

System Performance Validated

- System tested for usability (Fig-6); core pages loads quickly.
- Task-matching logic prioritizes urgency & availability correctly.

Architecture & Data Integrity Confirmed

- Implemented system matches High-Level & Low-Level designs (Fig-2, Fig-3).
- All database relations (Fig-4) work correctly; test cases (Fig-5) passed.

Social Impact

- Connects people for daily assistance, improving well-being and reducing stress.
- Creates job opportunities and improves access to essential services.
- Reduces social isolation and supports health with timely reminders.
- Fosters community support and strengthens connections among individuals.
- Enhances overall quality of life through timely help and reliable services.

Conclusion

- Strengthens communities by providing timely assistance and creating job opportunities.
- Enhances well-being, accessibility, and safety for all users.
- Fosters connections, reduces isolation, and improves overall quality of life.

References

- Cohen, S., & Wills, T. A. (1985). Stress, social support, and the buffering hypothesis. Psychological Bulletin, 98(2), 310-357.
- Berkman, L. F., & Glass, T. (2000). Social integration, social networks, social support, and health. Social Epidemiology, 1, 137-173.